

A UPF-Driven Multi-Voltage Three-Chiplet SoC with DVFS, Carried from RTL to GDSII

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Abstract—We present a three-chiplet system-on-chip carried from Unified Power Format intent to GDSII on the Nangate 45 nm open cell library. An application processor, a neural processing unit built around a 4x4 weight-stationary systolic array, and a shared L2 exchange fixed 54-bit flits over a three-port network-on-chip. A single IEEE 1801 description defines four power domains, of which the NPU domain is switchable across three supply voltages and full power gating. Because the NPU also runs on its own scaled clock, its interface is at once a clock-domain crossing and a supply crossing, and the isolation, level shifting and resynchronization it requires are inserted by the tools from that intent rather than by hand. The design closes at +3.985 ns worst slack on a 580 x 580 um die of 66,016 standard cells, holds static IR drop to 31 mV against a 47.5 mV budget, and reaches 99.06% transition-fault coverage. Power extracted from the routed database falls 63% between the full and quarter bins, while energy per cycle rises 45%, quantifying the efficiency cost of lowering the operating point.

Index Terms—UPF, IEEE 1801, multi-voltage design, DVFS, power gating, clock-domain crossing, network-on-chip, systolic array, physical design.

I. INTRODUCTION

Transistor density has continued to improve from one process generation to the next, but supply voltage and energy per operation have not kept pace [1]. A modern die therefore holds more logic than can be operated at once inside its power envelope, and the difference is recovered by running parts of the chip at reduced voltage or switching them off when idle. That is as much a physical design problem as an architectural one: every boundary between two power states needs isolation and level shifting, and if the two sides run on unrelated clocks, resynchronization as well.

This paper presents a three-chiplet system-on-chip built around that boundary and carried through a complete RTL-to-GDSII flow on the Nangate 45 nm open cell library [2]. An application processor, a neural processing unit and a shared L2 communicate over a network-on-chip. Only the NPU is bursty enough to be worth scaling and gating, so only it receives a switchable supply and a runtime-selected clock, concentrating the difficulty at a single interface (Fig. 1). An accelerator of this kind is idle for much of a workload, which is what makes the switchable domain worth its overhead: the header ring, isolation and level shifters cost area and delay at one boundary only, while the block they gate is the largest on the die.

The contributions are threefold. First, a four-domain multi-voltage implementation in which every isolation, level-shifter and header cell is inserted by the place-and-route tool from one IEEE 1801 description, so the intent that is verified is the intent that is built. Second, a boundary that carries both

crossings at once, verified under power-aware simulation with an assertion layer and monitored at runtime by a metastability canary. Third, post layout characterization at each DVFS operating point, taken from the routed database rather than scaled from synthesis, which shows both what voltage and frequency scaling saves and where it stops helping.

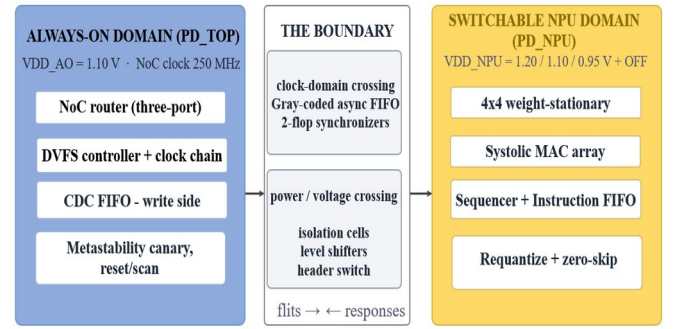


Fig. 1. The NPU interface is a clock-domain crossing and a supply crossing at the same time.

II. BACKGROUND AND RELATED WORK

A. Interconnect and clock-domain crossing

Once blocks are independently clocked and powered, a shared bus becomes the bottleneck, since one arbiter serializes every transfer, so the interconnect routes packets rather than wires [3]. This design uses one three-port router and a single 54-bit flit carrying a 2-bit destination, source and opcode, a 16-bit address and 32 bits of data (Fig. 2), which keeps the boundary logic uniform.



Fig. 2. The 54-bit flit. One format carries every transaction; the example is the AP writing to the L2.

Two flip-flops on unrelated clocks cannot exchange multi-bit data directly, since a signal sampled near an edge can become metastable [4]. Control bits cross through two-flop synchronizers and data through an asynchronous FIFO [5]. The pointers must cross as well, which is why they are Gray coded: consecutive values differ in a single bit, so a pointer sampled mid-transition is at worst one step stale rather than a value that never existed. Full and empty are then computed conservatively from the synchronized copies, making overflow and underflow impossible by construction

(Fig. 3). The same FIFO reports the occupancy that drives frequency selection.

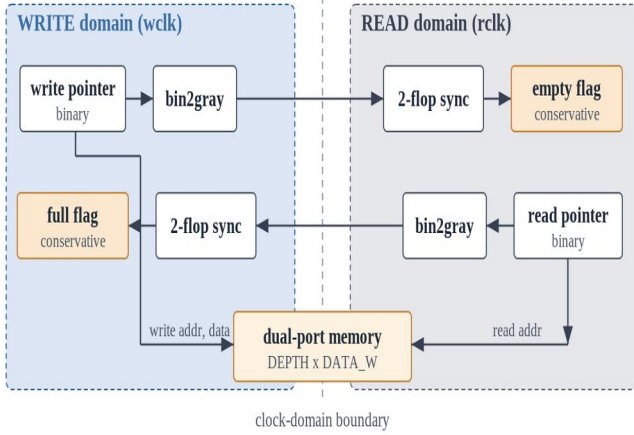


Fig. 3. Asynchronous FIFO. Only Gray-coded pointers cross the boundary, and the flags are computed from the synchronized copies.

B. Voltage and frequency scaling

Switching power in a CMOS block follows

$$P = \alpha CV^2f \quad (1)$$

so reducing the supply reduces power quadratically while reducing the clock reduces it linearly [6]. Frequency alone is weaker for a second reason: leakage is unaffected by it, and a slower block leaks for longer, so energy per computation can increase. Useful savings therefore require the supply to move with the clock, and idle periods to be gated rather than merely slowed. Fig. 4 shows this at synthesis with the supply held fixed; the leakage there is a synthesis estimate, lower than the post-route values in Section VI.

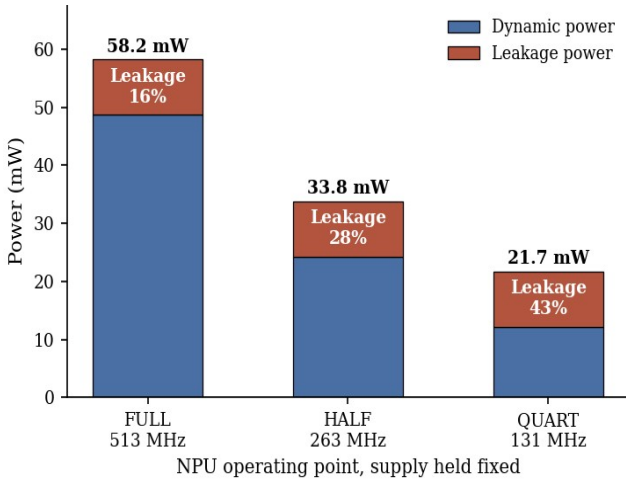


Fig. 4. Frequency-only scaling with the supply held fixed. Leakage does not scale, so its share of total power grows as the clock slows.

C. Multi-voltage cells and power intent

Four cell types make a switchable domain safe [7]. Header switches gate the domain supply, level shifters rescale signals crossing between different voltages, isolation cells clamp the outputs of a powered-down domain, and retention registers hold state on an always-on supply. The Unified Power Format, IEEE 1801 [8], describes this intent

separately from the RTL, so one file drives both power-aware simulation and physical implementation.

D. Related work

Physical low-power implementation on academic infrastructure has been reported mainly for single processor cores. Jimenez et al. applied static multi-supply voltage and power gating to a RISC-V processor in 22 nm FDSOI, where voltage islands alone returned roughly 3% of total power [9]. Munnangi and Stillmaker compared coarse- and fine-grained power gating on a five-stage RISC-V core in the same Nangate 45 nm library, fine-grained gating reducing post-layout power by 46% at negligible area cost [10]. Gowda et al. combined synthesis- and placement-stage optimizations in 130 nm for a 46% power and 11% area reduction, though multi-voltage and power gating were unavailable in their library [11]. Ravish Aradhya et al. reported a full RTL-to-GDSII RISC processor without multi-voltage intent [12]. These gate or optimize one processor at a fixed supply. This work operates four domains, one switchable across three voltages, and places the clock-domain and supply crossings on the same interface. The compute core follows Kung [13] and the parallel multiply-accumulate organization of inference accelerators [14].

III. SYSTEM ARCHITECTURE

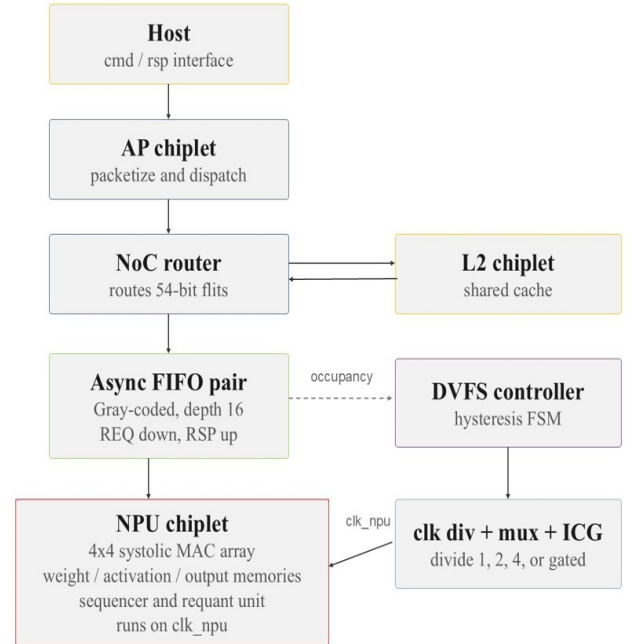


Fig. 5. The three chiplets. Traffic to the NPU passes through the FIFO pair, whose occupancy drives the DVFS controller.

A host issues transactions to the AP, which packetizes each into a flit and injects it into the router (Fig. 5). The L2 is reached directly, the NPU across the clock-domain crossing through a request FIFO and a mirror-image response FIFO. The AP also holds the power-management registers driving the NPU power sequence, so the sequence is software-visible.

Within the NPU, an instruction FIFO feeds a sequencer executing a short load-and-compute instruction set. The array

is 4x4 and weight-stationary, evaluating a matrix-vector product in rows + columns - 1 cycles. Each processing element skips the multiply when either operand is zero, so on sparse data the multiplier does not toggle and the saving is switching energy, not cycles alone. A requantization stage returns the wide accumulations to 8 bits through bias, activation, scaling and saturation.

The L2 serves packetized reads and writes from a 2 KB flip-flop array, since the library provides no compiled memory. Its write counter contains a 32-bit ripple-carry chain, and that chain rather than the systolic array sets the critical path. The DVFS controller is a four-state machine on the always-on clock that observes request-FIFO occupancy, using asymmetric watermarks and a minimum dwell time so it cannot oscillate between adjacent bins, and drives a programmable divider, glitchless multiplexer and clock gate. A metastability canary passes a known pattern through the same synchronizer and counts mismatches, so the crossing is monitored rather than assumed correct.

IV. POWER ARCHITECTURE AND INTENT

The design has four power domains. PD_TOP is always on at 1.10 V and holds everything that must survive an NPU power-down: the router, both FIFOs, the DVFS controller, clock generation, reset and scan. PD_NPU is switchable across 1.20, 1.10 and 0.95 V plus full shutoff through a header ring, being the largest and busiest block and so holding the most reclaimable idle energy. PD_AP and PD_L2 run at a fixed 1.10 V but are fenced separately so each is isolated and verified independently.

One UPF file declares the domains and supplies, the software-controlled header switch, clamp-to-zero isolation on the switchable domain outputs, level shifting in both directions across its boundary, and a power-state table of the legal system states. Each strategy maps to a named library cell, which is what lets the tool insert the cells during implementation (Fig. 6).

```
create_power_domain PD_NPU -elements {u_npu}

create_power_switch sw_npu -domain PD_NPU
-input_supply_port {vin VDD_A0}
-output_supply_port {vout VDD_NPU}
-control_port {ctrl u_ap/npw_pwr_on}
map_power_switch sw_npu -lib_cells {HEADER_X4}

set_isolation iso_npu -domain PD_NPU
-applies_to outputs -clamp_value 0
map_isolation_cell iso_npu
-lib_cells {ISO_FENCE0}

set_level_shifter ls_npu -domain PD_NPU
-applies_to both -rule both
map_level_shifter_cell ls_npu
-lib_cells {LS_LH LS_HL}

add_port_state VDD_NPU -state {HIGH 1.20}
-state {NOM 1.10} -state {LOW 0.95}
-state {OFF off}
```

Fig. 6. Power intent for the switchable NPU domain.

V. IMPLEMENTATION

A. Functional and power-aware verification

The RTL comprises 22 synthesizable modules. A 23-test self-checking regression passed before any power intent was applied, checking specific results rather than activity alone: a

matrix-vector product against its golden vector, zero-skip bypassing 122 of 128 multiplies on sparse data, residency across 13 DVFS bin transitions, and a bit-exact accumulator across the crossing. During the stress test the canary recorded 19 events with no data corrupted, showing the boundary was exercised and the synchronizers absorbed it.

The suite was then run under IEEE 1801 semantics, so a powered-down region genuinely loses state and the isolation clamps take effect. An assertion layer of 114 instances across five modules evaluated 74,874 times without a violation, covering FIFO overflow and underflow, legal router destinations, legal bin selection, and mutually exclusive skip and multiply decisions in every PE. The power sequence was confirmed in waveform: isolation asserts before the rail gates, and supply is restored before isolation releases.

B. Synthesis and physical implementation

Synthesis with scan insertion placed the timing wall at 1.95 ns, or 513 MHz, at the fast corner. Place and route committed the power intent and floorplanned the four domains on a 580 x 580 μm die (Fig. 7). Fence sizes follow from each block's synthesized area divided by a target utilization below 70%, leaving room for routing and for the cells the tool inserts; achieved utilizations are 68.0, 53.1 and 20.4% for PD_NPU, PD_L2 and PD_AP. The switchable domain occupies a corner, where its header ring and dedicated rail are simplest to build.

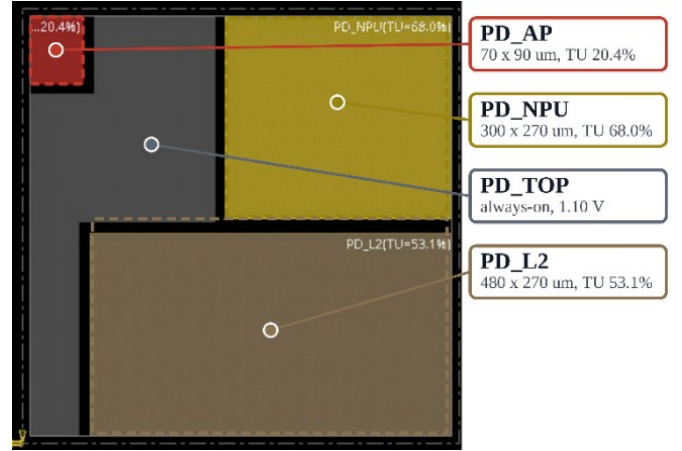


Fig. 7. Power-domain floorplan, with always-on channels separating the three fenced domains.

The power grid was sized from the sign-off current rather than chosen by inspection. With P the power drawn at the lowest supply, one stripe of sheet resistance R_s , length L and width W , and N stripes in parallel, the drop across the grid is

$$\Delta V = \frac{P R_s L}{V W N} \quad (2)$$

For $P = 20.9 \text{ mW}$ at $V = 0.95 \text{ V}$, $R_s = 0.035 \Omega$ per square, $L = 560 \mu\text{m}$ and $W = 1 \mu\text{m}$, eleven stripes per direction is the smallest grid that meets the 5% budget of 47.5 mV, at 39 mV. The implemented grid uses a 40- μm pitch, fourteen stripes per direction, giving 31 mV. A ring of 48 header switches encloses the switchable domain, and placement legalized

66,016 standard cells together with 299 multi-voltage cells inserted from the power intent. Detailed routing completed with no overflow.

One detail is worth recording for anyone reproducing this flow. The isolation, level-shifter and header cells are dual-rail while the placement rows are single-rail, so the row supply rail passes over the second supply pin of each such cell. The rail is split at those crossings as a post-route change, after which physical verification is clean and power-domain verification confirms correct connectivity in every domain.

VI. RESULTS

All 23 self-checks pass under both simulators. The DVFS workload is two bursts of thirty NPU instructions separated by a 200-cycle idle interval, which fills the request FIFO, drains it and fills it again, driving the controller through its operating points. Functional coverage of the bin covergroup reached 79.17%. The uncovered bin is an artifact of the covergroup sampling the two-bit clock-select output, which encodes only three distinct values, rather than the controller state; the residency counters confirm that all four operating points are entered.

TABLE I
POST-ROUTE POWER AT EACH OPERATING POINT

Bin	Freq.	Corner	Power	pJ/cycle
FULL	513 MHz	1.25 V	52.26 mW	102
HALF	263 MHz	1.10 V	29.35 mW	112
QUART	131 MHz	0.95 V	19.33 mW	148
OFF	gated	off	-	-

Table I reports power extracted from the routed database with the clock set to each bin in turn, so the tool computes the dynamic component directly. Each bin is evaluated at the library corner nearest its declared supply, so the full bin appears at 1.25 V rather than the 1.20 V UPF state. Total power falls 63% between the full and quarter bins, and leakage falls 85%, reducing its share from 35 to 15%. Energy runs the other way: energy per cycle rises from 102 to 148 pJ, because the remaining leakage integrates over a longer period. Scaling is therefore a power decision rather than an efficiency one, and the quarter bin pays only when the workload tolerates the added latency (Fig. 8).

Timing closed at +3.985 ns worst slack with no violating paths, physical verification is clean, and power-domain verification reports correct power and ground connectivity across all four domains (Fig. 9). Re-optimizing the clock tree on the same placed database cut buffers from 507 to 102 over the same 9,457 sinks, lowering clock-network power from 2.17 to 1.67 mW while holding +4.61 ns of slack. Transition-fault ATPG reached 99.06% coverage using 7,524 patterns.

These mechanisms are the ones production systems rely on: mobile parts gate an idle accelerator and scale its supply under load, and safety-critical devices rely on isolation and level shifting so one block fails cleanly when a neighboring domain is powered down.

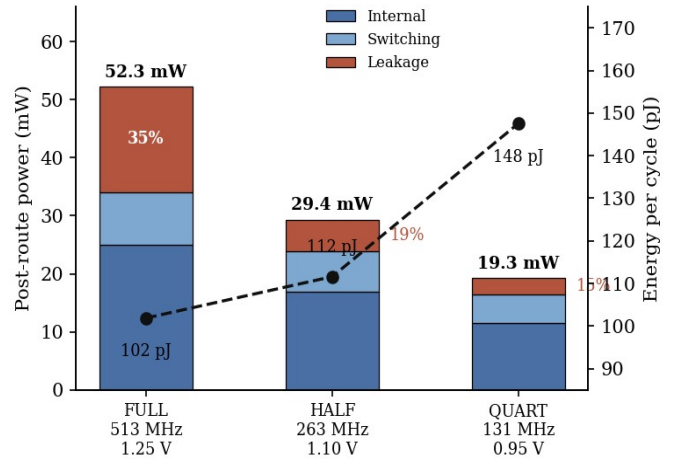


Fig. 8. Post-route power and energy per cycle at each operating point.

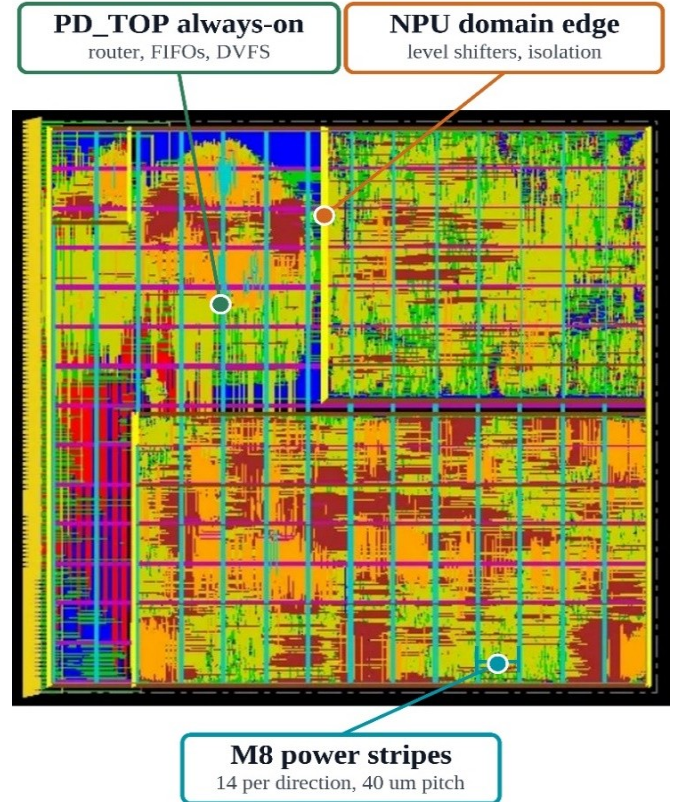


Fig. 9. Final routed die with the four power domains and their always-on channels.

VII. CONCLUSION

A four-domain multi-voltage system-on-chip was carried from a single power-intent description to a signed-off physical database: 299 isolation, level-shifter and header cells placed by the tool, timing closed at +3.985 ns, the grid within its 31 mV budget, and 99.06% transition-fault coverage. Power at each operating point shows a 63% reduction across the bins with a 45% rise in energy per cycle, the quantity a runtime policy must trade. Extensions include a compiled L2 memory, retention registers, and vectored rail analysis.

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